

VENIGALLA SHAMANTH CHOWDARY

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PROFESSIONAL SUMMARY

Computer Science and Engineering (Data Science) student skilled in Python, Java, SQL, machine learning, data analytics, and full-stack development. Experienced in building AI-powered applications using RAG, FastAPI, Flask, React, Streamlit, and geospatial technologies through internships, academic projects, and hackathons. Interested in software development, data science, and AI-driven solutions.

EDUCATION

Bachelor of Technology in Computer Science and Engineering (Data Science)

Rajeev Gandhi Memorial College of Engineering and Technology, Nandyal

2023 – 2027

CGPA: 8.52/10

Intermediate (MPC)

Narayana Junior College, Nandyal

2021 – 2023

Score: 97.5%

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL

Frontend Technologies: HTML, CSS

Backend Technologies: Flask, FastAPI, REST APIs

Databases: PostgreSQL, MySQL, SQLite

Machine Learning & AI: scikit-learn, pandas, NumPy, XGBoost, Prophet, RAG, FAISS, Sentence Transformers, Ollama

Developer Tools: Git, GitHub, GitLab, Docker, Streamlit, Jupyter Notebook, VS Code, Google Colab

Data Visualization: Matplotlib, Seaborn, Plotly, Tableau

PROJECTS

TravelSathi – Multilingual AI Cultural Travel Companion | [GitHub](#) | [Live Demo](#) | 2026

Tech Stack: Python, Streamlit, RAG, FAISS, Sentence Transformers, Ollama, Gemini, Groq

- Built an AI-powered travel companion supporting **3 languages — English, Telugu, and Hindi** — for localized cultural and travel assistance.
- Implemented **Retrieval-Augmented Generation (RAG)** using FAISS and semantic search to provide context-grounded travel and cultural responses.
- Developed **3 AI modules** covering personalized trip planning, festival exploration, and local phrase assistance.
- Integrated **offline Ollama inference** and BYOK support for cloud AI providers including Gemini, Groq, and OpenRouter.

SafePath Hyderabad – AI-Powered Nighttime Route Safety Platform | [GitHub](#) | [Live Demo](#) | 2026

Tech Stack: Python, FastAPI, React, Vite, GeoPandas, Leaflet, Gemini, Docker

- Developed a nighttime route-planning platform evaluating routes across **5 safety factors**: illumination, police proximity, medical access, pharmacy availability, and transit connectivity.
- Designed a weighted safety index using **50% illumination, 20% police proximity, 15% medical access, 10% pharmacy availability, and 5% transit connectivity**.
- Integrated geospatial datasets with GeoPandas and implemented route caching, address handling, deterministic building offsets, GPS navigation, and interactive maps.
- Added a **Gemini-powered route-specific AI safety assistant** and deployed the full-stack application using Docker, Render.

INTERNSHIP EXPERIENCE

AI Engineer Intern | Viswam AI, Swecha — Summer of AI Program | May 2026 – Present

- Lead data collection and preprocessing for a **Telugu Automatic Speech Recognition (ASR)** model, preparing audio-text training corpora for model development.
- Developed a **Retrieval-Augmented Generation (RAG)** system for question answering over Telugu books using semantic retrieval.
- Engineered features and resolved issues across **open-source AI applications**, improving functionality and reliability.
- Contributed to Telugu ASR development by preparing datasets, supporting model training, and improving model performance.
- Guided multiple intern teams on project tasks and priorities while streamlining communication and status updates with project managers.

LEADERSHIP AND ACHIEVEMENTS

Vice Chair, IEEE Information Theory Society, RGM CET — contributed to organizing technical activities and student initiatives.

2nd Runner-up – Best Startup Innovation, Ignite 8.0 (AIC-SK University)

2nd Place — CSE Data Science Poster Presentation, RGM CET.

CERTIFICATIONS

Automation Developer Associate — UiPath | Jan 2026 | [Certificate](#)